

their phones into 12.7cm (7-inch) tablets. LG is also said to have come out with a smartphone that folds outside to form a tablet. Patent drawings represent a conceptual LG phone that has a foldable display which can be used even after bending. Apple is considering the use of bendable displays on the iPhone in 2018.

Potential applications

Other technologies could also benefit from flexible displays. For example, flexible displays could be used as clothing that changes colour or pattern instantly as per the environment, making these particularly useful for soldiers.

Phones can also take a new format, similar to the Lenovo Cplus prototype, which starts as a thin handset but folds into a wristband. Samsung experimented with the introduction of curved displays on its original Galaxy Gear S smartphone, but moved to a more traditional flat, rounded Galaxy Gear S3. With curved displays, intelligent observers can theoretically have massive screens that surround their wrist to display even more information at once.

Cars could also use curved screens as more and more dashboards are going digital. For example, Mercedes E-Class displays infotainment indicators and infotainment data on a 31cm (12.3-inch) HD display that extends on the dashboard. Tesla and X models have a huge 43cm (17-inch) central display, but you can use a curved screen to make the outline look natural on the car's dashboard.

The biggest challenges

The concept of bent displays is not new, because gadget manufacturers appear to be displaying them on large shows like CES (Consumers Electronics Show in Las Vegas, USA) and MWC (Mobile World Congress in Barcelona, Spain) annually, but no one has brought a flexible display to the market yet. Why?

Since bending the display is difficult, crafting a device that uses one effectively is even more difficult. In order to create a flexible product, manufacturers need to know how to make each component flexible. Batteries are particularly difficult to bend because current technology requires batteries to keep their shape. After being bent, batteries are short-circuited and, if shorted too far in the housing, can explode.

Both LG and Samsung have been working on a flexible battery for many years, but the technology is still not available. So far, the only way to use a curved display is to make a device that bends on the hinge. This concept works with the Lenovo Yoga Book, which is equipped with a touch screen and a touch pad that doubles as a drawing tablet. With the hinge, device manufacturers can potentially use fixed batteries and silicon, but add a fold to them.

We are still waiting for the first device to hit the market, but when it arrives, we can expect durability and endless possibilities in design and innovation. Over the last few years, the patterns of phones and tablets have been outdated, and flexible displays can make them exciting again. **EFY**

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Binay Opto Electronics Pvt. Ltd.

27, PANDITA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
email: info@binayled.com, http://www.binayled.com

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